



PRODUCT DATA SHEET

SEMI SYNTHETIC CUTTING OIL – TECHNOFLUID SEMI SYNTHETIC CUTTING OIL

Product Description

- TECHNOFLUID Semi Synthetic Cutting Oil is a high-performance metalworking fluid formulated using a balanced blend of mineral oils, synthetic components, and advanced additive technology.
- Designed to deliver excellent cooling, lubrication, and cleanliness during machining operations.
- Forms a stable, translucent emulsion when mixed with water, providing clear visibility of the tool and workpiece.
- Offers superior resistance to bacterial growth and emulsion breakdown, ensuring long sump life.
- Free from nitrites, phenols, and other harmful substances, making it operator-friendly.

Key Performance Benefits

- Excellent cooling and lubricating properties improve tool life and machining efficiency.
- Stable emulsion provides consistent performance over extended operating periods.
- Superior anti-wear and EP performance supports medium to heavy-duty machining operations.
- Effective rust and corrosion protection for machined parts and machine components.
- Low foaming tendency ensures smooth operation in high-pressure coolant systems.
- Reduced misting and low odor improve shop-floor working conditions.
- Helps reduce overall coolant consumption and maintenance costs.

Applications

Recommended for a wide range of machining operations including:

- Turning, milling, drilling, boring, and tapping
- Grinding operations

Suitable for machining of:

- Mild steel and carbon steel
- Alloy steels
- Cast iron
- Selected non-ferrous metal.

Compatible with CNC machines and conventional machine tools.

Typical Uses

- General-purpose and precision machining operations requiring balanced cooling and lubrication.
- Centralized coolant systems and individual machine sumps.
- Applications demanding long fluid life and consistent machining performance.

Operating Characteristics

- Forms a stable emulsion at recommended dilution ratios.
- Maintains pH balance and corrosion protection over long service intervals.
- Compatible with commonly used machine tool materials and seals.

Typical Characteristics**

Characteristics	Test Method	Typical Figures
Appearance (Concentrate)	Visual	Clear / Light Amber
Appearance (5% Emulsion)	Visual	Translucent / Light Milky
Base Fluid Type	—	Semi-Synthetic (Mineral + Synthetic)
Emulsion Stability (5% in water)	IP / In-house	Stable
Density @ 15°C, g/ml	ASTM D4052	0.880
Kinematic Viscosity @ 40°C, cSt	ASTM D445	30 – 40
pH of 5% Emulsion	ASTM D1287	8.8 – 9.2
Refractometer Factor	—	2.5 – 3.5
Flash Point (COC), °C	ASTM D92	≥ 180
Pour Point, °C	ASTM D97	-6
Copper Corrosion (3 hrs @ 100°C)	ASTM D130	Pass

Rust Prevention (Chip Test)	ASTM D4627	Pass
Foam Tendency	ASTM D892	Low
Hard Water Stability	In-house	Good
Biostability	In-house	Excellent
Operating Dilution Ratio	—	5% – 10%